Dumindu Kumarapeli

+94 76 085 8628 | duminduku.25@gmail.com | Colombo, Sri Lanka
linkedin.com/in/dumindu-kumarapeli-7a4636309 | github.com/Dumee-25 | portfolio.kadpkumarapeli.workers.dev

Data Science undergraduate building LLM systems end to end (RAG, agents, local-model inference), with first-author research on privacy-preserving LLM analysis, epidemic forecasting and clinical prediction.

Education

NSBM Green University — Sri Lanka
BSc (Hons) Data Science, currently Year 3 — *2025 – Expected Dec 2028*

- GPA 3.85/4.00.

Research and Publications

- **D. Kumarapeli** and H. Weerasinghe, "Privacy-preserving automated exploratory data analysis using deterministic summarization and local large language models," *Proc. NSBM Student Research Symposium on Innovation Technology (NSRSIT)*, 2026, pp. 146–150. **Best Overall Research**, session award.
- **D. Kumarapeli** and H. Weerasinghe, "Timescale separation and model reduction in operational dengue forecasting," ICCMM 2026, University of Colombo. [accepted, oral presentation]
- **D. Kumarapeli** and H. Weerasinghe, "A regime audit for detecting constraint-artefact reproduction numbers in hybrid mechanistic–machine learning epidemic models." [under review]
- **D. Kumarapeli** and H. Weerasinghe, "A hybrid compartmental and neural network model for district-level dengue forecasting in Sri Lanka: forecast skill, spatial heterogeneity and identifiability." Target: *Ceylon Journal of Science*. [in preparation]
- **D. Kumarapeli** and H. Weerasinghe, "Proxy labels versus observed endpoints: an audit and benchmark of clinical benefit prediction on MSK-CHORD." [in preparation]

LLM and AI Projects

DataSense: Privacy-Preserving Automated EDA | *FastAPI, Polars, Ollama* — github.com/Dumee-25/DataSense

- Designed a seven-step async pipeline that computes the statistics first and sends only the resulting facts to the LLM, so raw rows never reach the model; this is the method in the NSRSIT paper.
- Made the LLM layer pluggable (local Ollama by default, OpenAI or Groq optional) with caching and a rule-based fallback; output covers quality, outlier and Simpson's-paradox checks and a model recommendation.

OWL: Datacenter AI Operations Platform | *Gemini, MCP* — github.com/Dumee-25/OWL-Datacenter_agent

- Built an agentic RAG assistant over ChromaDB with hybrid vector and keyword retrieval and recency re-ranking, served through REST, WebSocket and a nine-tool MCP server.
- Paired an Isolation Forest and LSTM autoencoder ensemble with Gemini-generated root-cause reports; every alert carries a confidence score and the system never acts on infrastructure.
- Wrote a simulator producing 4.7M telemetry rows over 18 months; logged per-call LLM latency and cost.

Fulyn: Private Tool-Calling Agent | *pgvector, Ollama* — github.com/Dumee-25/Fulyn

- Agent turns free-text diary entries into linked expense, sleep, mood and journal records; every tool call is schema-checked and invalid arguments are returned to the model to retry.
- Replaced vector-only search after finding unrelated queries scored 0.62 similarity, higher than some true matches; now merges vector and full-text rankings and flags keyword-free hits.
- Ran opt-in tool-calling tests across local models; llama3.2 failed most cases by picking the wrong tool or inventing arguments.

Technical Skills

LLM / RAG: Retrieval-augmented generation (RAG), AI agents and tool calling, prompt engineering, Ollama, Gemini, OpenAI API, ChromaDB, pgvector, Model Context Protocol (MCP)
ML / Data: PyTorch, TensorFlow, scikit-learn, XGBoost, pandas, Polars, NumPy, SciPy, YOLOv8, OpenCV
Languages: Python, SQL, Java, JavaScript, TypeScript, R
Backend / Tools: FastAPI, PostgreSQL, SQLite, Docker, Git/GitHub, Linux, Next.js, React, CUDA
Spoken: English (fluent), Sinhala (native)

Awards

Best Overall Research, session award, NSRSIT 2026 (2026) | Data Storm 7.0 Final Round, Top 10, team Data Drifters (2026)